

Journal: *Ecology*

Manuscript type: Article

Manuscript ID: ECY25-1549

**Environmental and spatial heterogeneities shape aquatic communities at tributary
confluences in a desert river network**

Jared W. Freedman^{1*}, Theodore A. Kennedy², Molly K. Burke¹, David A. Lytle¹

1. Department of Integrative Biology, Oregon State University, Corvallis, Oregon, USA

2. Grand Canyon Monitoring and Research Center, Southwest Biological Science Center, USGS,
Flagstaff, Arizona, USA

* Present address: Department of Natural Resources and the Environment, Cornell University,
Ithaca, New York, USA

Corresponding Author: Jared W. Freedman, jwf236@cornell.edu

Open Research Statement:

All sequencing data are available at the National Center for Biotechnology Information (NCBI)

Sequence Read Archive Data under accession number PRJNA1347436 (BioSample accession

numbers SAMN5258034–SAMN5258127) at

<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1347436/>. Supporting data and code (Freedman
et al, 2026) are available in Dryad at <https://doi.org/10.5061/dryad.tjq2bwf8>.

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25 Keywords:

26 Aquatic invertebrates; Colorado River; Community structure; Environmental DNA; Grand

27 Canyon; Landscape Ecology; Metacommunity ecology; River networks

Abstract

Landscape heterogeneities influence dispersal between habitats and environmental conditions within habitats, shaping the metacommunity dynamics that govern the distribution of species throughout an ecosystem. Riverine aquatic communities are sensitive to heterogeneity in flow, substrate, and other abiotic conditions arising from interactions between aquatic and terrestrial landscapes, as well as from spatial heterogeneity in river networks that can cause isolation and fragmentation. Landscape characteristics can be particularly variable at tributary confluences, where branches with different biotic and abiotic characteristics join together. Here, we investigate the effect of tributary confluences on aquatic invertebrate community connectivity in Grand Canyon, testing the competing hypotheses that a) tributary confluences facilitate community connectivity by linking habitats and inducing gradual shifts in habitat conditions, or b) differences in adjacent tributary and mainstem habitat conditions are sufficient to isolate tributary communities from nearby mainstem communities. Using invertebrate-specific environmental DNA (eDNA) metabarcoding, we quantified aquatic invertebrate community structure at 22 tributary confluences in Grand Canyon, with samples collected at the tributary mouths and in the mainstem Colorado River. We identified 667 aquatic invertebrate taxa in total, with assemblages dominated by Diptera, and tributary communities possessed higher taxonomic richness than mainstem communities. Community structure at confluences differed between tributary and mainstem habitats, with high beta diversity driven by species turnover rather than nestedness. Between tributary communities throughout Grand Canyon, beta diversity was again driven by high turnover, with environmental heterogeneity explaining more of the community structure than spatial relationships of the network. The significance of turnover at multiple scales suggests that changes in community composition are driven by local dynamics of species sorting

and habitat filtering, and that ecological isolation is an important dynamic in structuring aquatic invertebrate communities. Tributaries contribute to the high aquatic invertebrate diversity in Grand Canyon, but their isolation relative to each other leaves communities vulnerable to habitat degradation and stochastic disturbances. Together, these results suggest that landscape heterogeneity at tributary-mainstem confluences limits ecological connectivity even at short geographical distances, resulting in an archipelago-like pattern of disjunct tributary communities despite the hydrological connectivity within the river network.

Introduction

Although rivers, oceans, and terrestrial landscapes can form continuous habitats across the Earth's surface, landscape heterogeneity – the qualitative or quantitative variation in landscape elements – causes discontinuities in species distributions (Tonetti et al., 2023). Such heterogeneities can create abrupt changes in landscape characteristics that, whether natural or anthropogenic in origin, reduce connectivity between habitat patches and create barriers to dispersal and colonization that fundamentally alter ecological dynamics at multiple scales (Caplat et al., 2016; Gustafson and Gardner, 1996; Wang and Bradburd, 2014). Discontinuities in landscape structure are frequently the result of keystone abiotic processes, such as fires, climate gradients, or variable flow regimes (Holling, 1992). These influential processes shape key components of the landscape at multiple scales, and are often representative of other environmental and ecological variables (Holling, 1992; Nash et al., 2014).

In river networks, the hydrological, geomorphological, and environmental features of the drainage basin influence the magnitude of river flows, facilitating variation in flow regimes across time and space (Benda et al., 2004b; Melles et al., 2012; Vannote et al., 1980). The

strength and direction of river flows influence ecological connectivity in freshwater river networks (Peterson et al., 2013; Tonkin et al., 2018), and are expected to cause characteristic shifts in aquatic community structure and biodiversity as the network accumulates greater flows (Fagan, 2002; Vannote et al., 1980). However, the branching structure of the network can also impose breaks in ecological gradients where tributaries, lakes, or dams form discontinuities in the physical structure of the network (Benda et al., 2004b; Bruns et al., 1984; Ward and Stanford, 1983). Variations in flow regime and network connectivity related to landscape features can shape community structure through multiple, non-exclusive metacommunity dynamics (Brown et al., 2017; Brown and Swan, 2010; Leibold et al., 2004). For example, landscape discontinuities can abruptly alter environmental characteristics in aquatic habitats, creating distinct changes in community structure through localized dynamics of species sorting and habitat filtering (Heino et al., 2015; Suzuki and Economo, 2021). Alternatively, these discontinuous landscape features may create multiple pathways for dispersal across both aquatic and terrestrial landscapes (Campbell Grant et al., 2007), producing gradual shifts in community structure through regional dynamics of dispersal limitations, mass effects, and refugia effects (Heino et al., 2015; Mouquet and Loreau, 2003; Tonkin et al., 2018). Quantifying these metacommunity dynamics is therefore essential to determine the capacity of a heterogeneous landscape, such as a river network, to sustain biological diversity (Patrick et al., 2021; Suzuki and Economo, 2021).

Headwater tributaries commonly exhibit higher community heterogeneity than elsewhere in the river network (Meyer et al., 2007), yet the dynamics of aquatic communities at confluences of small tributaries and large mainstem rivers remain understudied. Large rivers are ecologically, economically, and socially important ecosystems (Nilsson et al., 2005), and tributary confluences in such systems can form discontinuities in river flows that alter

downstream environmental conditions and form nodes of connectivity between branches of the network (Benda et al., 2004b; Bruns et al., 1984; Kiffney et al., 2006). In mainstem habitats, confluences can induce shifts in community composition due to changes in habitat conditions (Milner et al., 2019) and downstream drift-mediated dispersal (Wilson and McTammany, 2014). In tributary habitats, on the other hand, lower river flows and diverse environmental characteristics can provide refugia from mainstem environmental conditions (Beckmann et al., 2005; Pracheil et al., 2013) and introduce additional dispersal pathways for tracking of optimal habitat conditions (Uno and Power, 2015). However, much previous ecological research at tributary confluences has focused on the effect of tributaries on taxa in the mainstem river (Beckmann et al., 2005; Czeglédi et al., 2016; Jones and Schmidt, 2017), and few studies have investigated the effects of tributary confluences with high flow asymmetry (i.e., where tributary flows are substantially lower than the mainstem; Jones and Schmidt, 2017; Milner et al., 2019).

The Colorado River network within Grand Canyon provides an ideal system to examine the influence of landscape heterogeneity at tributary confluences. The mainstem Colorado River is a large, seventh-order river with flow regulated by operations of Glen Canyon Dam, which creates modified flow conditions detrimental to many native freshwater organisms, including fish (Minckley, 1991), primary producers (Hall et al., 2015), and aquatic invertebrates (Miller et al., 2020). Notably, aquatic invertebrate communities of the Colorado River below Glen Canyon Dam are taxonomically limited and nearly devoid of insects due to environmental and niche filtering processes (Kennedy et al., 2016; Stevens et al., 1997). Invertebrate communities in the Colorado River are greatly reduced in abundance and diversity (Kennedy et al., 2016), while many low-order, free-flowing tributaries that enter the river support higher biodiversity (Oberlin et al., 1999; Whiting et al., 2014). These tributaries provide important ecological benefits to the

mainstem Colorado River, for example through stabilizing effects on food webs (Sabo et al., 2018) and maintaining connectivity of fish populations (Bouska et al., 2023). However, little is known about the ecological connectivity of aquatic invertebrate communities in Grand Canyon tributaries and how the composition of these communities is structured throughout the network. Limited precipitation and the elongate structure of the river network mean high-order rivers are often joined by low-order tributaries, leading to hydrological asymmetry between tributaries and the mainstem. The large mainstem Colorado River has annually consistent high flows with sub-daily variation due to dam operations, and environmental conditions (i.e., temperature, sediment load, dissolved oxygen) are homogenized by the impoundment of the Glen Canyon Dam (Webb et al., 1999). Free-flowing tributaries, on the other hand, possess much lower yet more variable flow regimes, and experience greater diversity in environmental characteristics such as temperature, benthic substrate, and water quality (Oberlin et al., 1999). This diversity in environmental conditions has the potential to shape diverse aquatic invertebrate communities through local metacommunity dynamics of environmental filtering (Brown and Swan, 2010). These differences between the Colorado River and its tributaries can alter the prevalence and severity of stochastic disturbances such as floods and droughts (Benda et al., 2004a; Jones and Schmidt, 2017), further influencing community dynamics at fine spatial scales (Bogan and Lytle, 2011). Furthermore, strong contrasts between the aquatic and terrestrial landscape and the potential for hydrological fragmentation increases landscape resistance to dispersal in arid river networks like Grand Canyon, making communities in such networks particularly sensitive to landscape heterogeneities (Bogan and Boersma, 2012; Morán-Ordóñez et al., 2015). In this case, communities in tributaries may function as isolated islands separated by a “sea” of inhospitable riverine and terrestrial terrain.

Here, we test competing hypotheses concerning the effects of tributary confluences on aquatic invertebrate communities in arid river networks. First, if tributary confluences facilitate community connectivity by linking habitats and inducing gradual shifts in habitat conditions, we should expect a nested community structure between proximal communities at tributary confluences and increasing biodiversity in the downstream direction owing to the gradual addition of tributary taxa. Alternatively, if environmental heterogeneity is sufficient to isolate tributary communities from nearby mainstem communities, we would instead expect tributaries to display divergent patterns of community turnover in comparison to the mainstem. To address these hypotheses at a large spatial scale, we utilize the efficiency of an environmental DNA (eDNA) approach to address a major management question in the Grand Canyon ecosystem, specifically whether unregulated tributaries serve as biodiversity reservoirs that can compensate for ecological degradation in the mainstem Colorado River.

Methods

Study sites and sample collection

In April 2022, we collected eDNA water samples from 22 paired mainstem Colorado River and tributary confluences along a 452 km segment of the Colorado River primarily in Grand Canyon (Figure 1; Appendix S1: Table S1). All perennial tributaries where access was feasible (i.e., safe and repeatable access to tributary banks and year-round surface flow into the Colorado River) were included in the study. At each confluence, tributary samples were collected from a riffle habitat upstream of the high-water influence of the Colorado River, while mainstem samples were collected from a location of positive flow directly upriver from the confluence of the tributary to prevent drift of eDNA from the tributary into the mainstem sample.

The paired sites were separated by 42-651 m, with an average distance of 174 m. Due to naturally inconsistent spacing of tributaries throughout Grand Canyon, supplementary non-paired mainstem samples were collected at river kilometer 32, 122, and 281. This sampling scheme resulted in a total of 47 locations where eDNA was collected, at 22 tributaries and 25 mainstem locations.

Four 1 L replicates were taken at each sampling location. For each sampling replicate, one liter of stream water was collected in sterile 1.5L roll-top bags (Whirl-Pak, Austin, TX, USA) while wearing nitrile gloves, and sampling locations were approached from downstream to prevent contamination between samples. Water samples were either processed immediately or stored in a cooler until filtration later in the same day (<8 hours). eDNA was obtained from water samples by filtering onto a 0.45 µm MCE filter (Sterlitech, Auburn, WA, USA) using a vacuum pump manifold. Filters were then preserved in 1.5 mL of ATL buffer (Qiagen, Germantown, MD, USA) and stored in an insulated container at room temperature until extraction. In addition to eDNA collection, environmental variables were recorded at each site, including GPS coordinates, water temperature, water dissolved oxygen, water pH, water conductivity, and substrate size. Additional environmental characteristics were obtained from StreamStats (USGS) for tributary habitats.

DNA extraction, sequencing, and bioinformatics

DNA extractions of preserved eDNA filters were performed using a modified protocol for the DNeasy Blood and Tissue kit (Qiagen; Appendix S1: Section S1). To increase DNA yield in each sample, we combined two filters from each site into a single extracted sample, resulting

in two 2 L replicates of extracted eDNA per site. Extracted DNA was then purified of PCR inhibitors using the One-Step PCR Inhibitor Removal Kit (Zymo Research, Irvine, CA, USA).

For each extracted replicate, we performed two PCR replicates using a two-stage PCR approach to amplify a diagnostic target region of the invertebrate mitochondrial COI gene (details in Appendix S1: Section S2). In the amplicon PCR stage, a 142 base pair fragment of the COI gene was amplified using the fwH2/EPTDr2n primer set (Leese et al., 2021). In the following library prep PCR stage, we used custom fusion primers to add dual unique indices and the Illumina adapter tails to the PCR amplicons from the first stage. PCR products were purified with the MinElute PCR cleanup kit (Qiagen). DNA concentration of PCR products was quantified using the Quant-iT platform (Thermo Fisher, Waltham, MA, USA) and equimolar amounts of PCR products were combined to create 50nM pooled libraries for samples from each year. DNA fragment length and quality were quantified using TapeStation (Agilent, Santa Clara, CA, USA). The eDNA library was sequenced by the Oregon State University Center for Quantitative Life Sciences on a single flow cell using the NextSeq 2000 150bp paired-end P1 kit (Illumina, San Diego, CA, USA).

Sequencing reads were demultiplexed using Illumina software, and the JAMP v0.77 metabarcoding pipeline was used to trim, filter and cluster reads into operational taxonomic units (OTUs) with <3% sequence similarity (Elbrecht et al., 2018). For the full metabarcoding pipeline, see Appendix SI: Section S3. OTUs were assigned taxonomic identification by aligning OTU sequences to the BOLD database using BOLDigger v2.0.4 (Buchner and Leese, 2020). OTUs with a reference similarity threshold < 85% were excluded from further analysis. Due to concordance in OTU composition (Appendix S1: Figure S1), reads from site and PCR replicates

were combined for further community analysis. The resulting dataset was transformed to presence/absence for community analysis.

Community Analysis

All analyses were carried out in R statistical software (version 4.5.1; R Core Team, 2025). Mean OTU richness, insect richness, and richness of Ephemeroptera, Plecoptera, and Trichoptera (EPT) taxa were compared between tributary and mainstem habitats using a Wilcoxon rank-sum test and effect sizes were quantified using Wilcoxon's r metric.

Nonmetric multidimensional scaling (NMS) ordinations with Sorensen distance were calculated using the R packages *vegan*, using 9,999 randomized iterations (Oksanen et al., 2020). Two PERMANOVA tests based on a Sorensen distance matrix with 9,999 permutations were used to test the influence of habitat type (tributary or mainstem) and river network distance on community dissimilarity (stream type model structure: community ~ stream_type; river network model structure: community ~ river_mile; Oksanen et al., 2020). PermDisp permutation tests with 9,999 pairwise iterations were used to test the dispersion of the community distances between tributary and mainstem sites (Anderson et al., 2006).

The relative importance of spatial and environmental variables on community composition in tributaries was tested with a variance partitioning procedure using distance-based redundancy analysis (dbRDA; Borcard et al., 1992). We performed this analysis only on tributary communities to further explore the mechanisms driving high beta diversity observed among this habitat type. To determine significant environmental predictor variables, forward selection of model parameters was performed with the *ordistep* function of *vegan* 2.6-4 (Oksanen et al., 2020). Significant environmental variables included temperature, drainage relief, 15th-

percentile substrate particle size, and percent barren terrain in drainage. Spatial distances between sampled habitats were calculated as the distance along the Colorado River between tributary confluences. Because tributary sampling locations were all located within 200 meters of the mainstem and the morphology of these confluences frequently changes flood disturbances (Webb et al., 1999), the distance from tributary confluence to tributary sampling location was not incorporated in river network distances. Next, we used the *varpart* function of *vegan* 2.6-4 (Oksanen et al., 2020) to calculate the variance in community structure (adjusted R^2) explained by spatial variation (river network distance), environmental variation (significant environmental variables), shared variation, and residual variation. The significance of explanatory variables on community dissimilarity was tested using ANOVA-like permutation tests with 9,999 iterations.

Analysis of modularity and nestedness in Grand Canyon aquatic invertebrate communities were performed on both a species co-occurrence network and a species-site network. Modularity measures the extent to which a network is partitioned into distinct, tightly interacting groups, while network nestedness measures the degree to which interactions represent hierarchical subset of species-poor groups nested within more diverse groups (Delmas et al., 2019). For species co-occurrences, a unipartite ecological network with nodes representing OTUs and edges representing positive interactions between OTUs was constructed using the R package *cooccur* (Griffith et al., 2016), retaining statistically significant positive co-occurrences ($p < 0.05$). Isolated nodes displaying no significant positive associations were removed prior to downstream analyses. Observed network modularity (Q) was calculated using the Louvain community detection algorithm (*cluster_louvain()*) in the R package *igraph* (Csárdi et al., 2026). The nestedness metric based on overlap and distance decreased fill (NODF) was calculated for the co-occurrence network with the *nestednodf()* function in the R package *vegan* (Oksanen et

al., 2020). To evaluate whether each of these metrics deviated from random expectations, a standard Z-score and empirical p -value were calculated by benchmarking each metric (Q or NODF) against a null distribution generated using a degree-preserving edge-swapping algorithm (*igraph::keeping_degseq*) on 999 randomized networks, which provides sufficient statistical power while balancing the high computational demands of the algorithm. To determine if modules in the co-occurrence network reflect taxonomic identity of member OTUs, we evaluated the global distribution of OTUs across modules using permutation-based Chi-square tests at the Order, Family, and Genus levels. P -values were calculated via Monte Carlo simulation using 9,999 permutations to account for the low-frequency rare taxa common in eDNA community datasets.

For the species-site interactions, a bipartite (two-mode) network was created that links OTUs to sites in which they occur. Modularity (Q) of this network was calculated from the presence-absence community matrix using the DIRTLPawb+ algorithm (*computeModules()*) in the R package *bipartite* (Dormann et al., 2009). This algorithm was replicated 10 times, and the iteration with the highest Q was selected as the observed empirical value. Nestedness was calculated for the site-species network using the NODF algorithm. Standard Z-scores and statistical significance of the observed Q and NODF for the species-site network were calculated by testing against a null distribution generated from 999 null matrices using the Curveball algorithm (*nullmodel()*) in the R package *vegan* (Oksanen et al., 2020), which retains constant site richness and OTU frequency across null matrices.

Patterns of beta diversity among tributary and mainstem sites were examined following the statistical framework of Baselga (2010), which partitions Sorensen dissimilarity (β_{sor}) into additive components of species turnover (β_{sim}) and nestedness-resultant dissimilarity (β_{sne}). This

framework, as implemented by the R package *betapart* 1.6 (Baselga and Orme, 2012), provides a quantitative method for identifying whether differences in biotic communities are driven by the gain and loss of species across spatial gradients (nestedness) or the replacement of species as a consequence of environmental or historical sorting (turnover). To evaluate the relative contributions of species turnover versus nestedness in driving dissimilarity at tributary confluences, β_{sor} , β_{sim} , and β_{sne} were calculated for each pair of tributary confluence and adjacent mainstem sites using the *betapart::beta.pair* function. Differences in β_{sim} and β_{sne} were compared using a paired Wilcoxon signed-rank test, and effect sizes were calculated using Wilcoxon's r metric. Additionally, to evaluate the role of turnover and nestedness-resultant dissimilarity throughout Grand Canyon, multiple-site β_{sor} , β_{sim} , and β_{sne} were calculated separately for three groups of communities: all sampled communities, tributary communities, and mainstem communities. To account for the sensitivity of multiple-site dissimilarity calculations to differing numbers of sites, mean multiple-site dissimilarities were calculated by subsampling sites at 90% of the smallest group size (tributary sites, $n = 22$) across 9,999 randomized iterations, using the *betapart::beta.multi* and *betapart::beta.sample* functions. Mean values of resampled β_{sor} , β_{sim} , and β_{sne} were reported and used for further analysis. Whether species turnover or nestedness-resultant dissimilarity was the dominant fraction of β_{sor} was determined using the $\beta_{\text{sim}} / \beta_{\text{sor}}$ ratio (β_{ratio}). Values of β_{ratio} range from 0 (complete dominance of nestedness-resultant dissimilarity) to 1 (complete dominance of turnover; Martín-Devasa et al., 2024). The significance of differences in beta diversity between tributary and mainstem habitats was calculated using a two-group permutation test that randomized site identities across 9,999 iterations.

Results

Alpha Diversity

We obtained an average of 302,353 reads per replicate (min=220,106, max=1,913,521) following the bioinformatic filtering in our metabarcoding pipeline. Across all sites sampled in 2022, we identified 667 aquatic-associated invertebrate OTUs with reference similarity >85% (Figure 2a). Alignment of OTU sequences with the BOLD reference database identified 161 species-level classifications, 138 genus-level classifications, and 299 family-level classifications. 497 OTUs had sequences aligning with the order Diptera, comprising 74.5% of all OTUs found in Grand Canyon. For the full list of detected taxa, see Freedman et al. 2026 (Taxa_List.csv).

Aquatic invertebrate communities in tributaries had significantly higher OTU richness compared to communities in the mainstem Colorado River (Wilcoxon rank sum $W = 471$, $p < 0.0001$, $r = 0.71$; Figure 2b). A total of 604 OTUs were found in tributary samples, with an average of 103 OTUs at each tributary site. In contrast, 232 OTUs were found in the mainstem, with an average of 47 OTUs at each mainstem site. 435 of the total 667 OTUs were found exclusively in tributaries, and 63 OTUs were found exclusively in the mainstem. Richness of taxa belonging to the class Insecta was higher in tributary communities than in mainstem communities (Wilcoxon rank sum $W = 78$, $p < 0.0001$, $r = -0.72$; Appendix S1: Figure S2), and Ephemeroptera, Plecoptera, and Trichoptera (EPT) taxa exhibited higher richness in tributaries communities than mainstem communities as well (Wilcoxon rank sum $W = 144.5$, $p < 0.0001$, $r = -0.47$; Appendix S1: Figure S3).

Among taxa identified at each tributary confluence (i.e., paired tributary and mainstem communities), $59.2 \pm 15.9\%$ (mean $\pm$ SD) of OTUs were found exclusively in the tributary community and $24.1 \pm 14.1\%$ of OTUs were found exclusively in the mainstem community

(Appendix S1: Figure S4). Taxa shared between adjacent tributary and mainstem communities comprised $16.7 \pm 4.7\%$ of the total taxa identified at each site pair.

Beta diversity at tributary confluences

Mean Sorensen dissimilarity was high for all comparisons of paired tributary and mainstem sites (mean $\beta_{\text{sor}} = 0.695$; Figure 3). This community dissimilarity was driven by high spatial turnover (mean $\beta_{\text{sim}} = 0.516$), while nestedness was significantly lower (mean $\beta_{\text{sne}} = 0.178$; Wilcoxon signed-rank $V = 249$, $p < 0.001$). The effect size of this comparison was large ($r = 0.97$), indicating that the distinct invertebrate communities observed across Grand Canyon tributary confluences were primarily driven by compositional differences between mainstem and tributary habitats, with few shared taxa between adjacent mainstem and tributary sites.

Beta diversity throughout network

Aquatic invertebrate community structure across Grand Canyon aquatic habitats was visualized using a two-dimensional NMS ordination of the presence-absence community dataset. The ordination closely fit the observed community structure (metric $R^2 = 0.881$, stress 0.1239; Figure 4), indicating that it was a strong representation of the community structure of the observed OTUs. The structure of invertebrate communities differed significantly between tributary and mainstem sites (PERMANOVA $R^2 = 0.250$, $F = 14.99$, $p < 0.0001$), and differences in beta diversity among tributary communities was significantly higher than among mainstem communities (PERMDISP $F = 47.19$, $p < 0.0001$). Position within the river network influenced community composition as well, with community dissimilarity significantly covarying with river mile (PERMANOVA $R^2 = 0.066$, $F = 4.24$, $p = 0.0051$).

Ecological network analysis revealed patterns of high modularity and low nestedness in Grand Canyon aquatic invertebrate communities. The species co-occurrence network (Appendix S1: Figure S5), which connects OTUs that co-occur more frequently than expected by chance, was highly modular ($Q = 0.52$, $Z = 174.58$, $p < 0.001$) and was not significantly nested ($NODF = 9.32$, $Z = -0.94$, $p = 0.941$). There was no significant pattern of taxonomic distributions across the observed modules when analyzed with permutational Chi-square tests (9,999 replicates) at the level of Order ($p = 0.298$), Family ($p = 0.292$), or Genus ($p = 0.305$). Analysis of the bipartite species-site network revealed similar patterns, with statistically significant evidence for network modularity ($Q = 0.35$, $Z = 4.99$, $p < 0.001$) but not nestedness ($NODF = 24.48$, $Z = 0.98$, $p = 0.282$). These network analyses indicate that aquatic invertebrate communities are highly compartmentalized, with taxonomically diverse groups of specialist species that co-occur alongside each other and relatively few backbone species found throughout the ecosystem.

Across all invertebrate communities in Grand Canyon, multiple-site Sorensen dissimilarity was high (mean $\beta_{\text{sor}} = 0.95$, resampled $n = 20$), and this diversity was primarily driven by turnover rather than nestedness-resultant dissimilarity ($\beta_{\text{ratio}} = 0.91$, resampled $n = 20$). When communities were grouped by habitat type (tributary and mainstem), multiple-site Sorensen dissimilarity was high for both types, and tributary habitats (mean resampled $\beta_{\text{sor}} = 0.90$, resampled $n = 20$) displayed significantly higher dissimilarity than mainstem habitats (mean resampled $\beta_{\text{sor}} = 0.85$, resampled $n = 20$; permutation test, $p < 0.001$). The proportion of dissimilarity derived from turnover was high among tributary habitats (mean $\beta_{\text{ratio}} = 0.93$, resampled $n = 20$) and among mainstem habitats (mean $\beta_{\text{ratio}} = 0.92$, resampled $n = 20$). High proportions of turnover indicate that invertebrate communities in these habitat groups are primarily structured by compositional turnover rather than nestedness-resultant dissimilarity.

Variance partitioning using dbRDA identified both environmental and spatial variables as significant in explaining aquatic invertebrate community structure among tributaries (Table 1). Environmental heterogeneity, composed of the significant variables of temperature, drainage relief, 15th-percentile substrate particle size, and percent barren terrain in drainage, explained 19.6% of observed variance in community diversity, while spatial relationships along the river network between tributary habitats explained 4.7% of community diversity, supporting the hypothesis that tributaries support more diversity than mainstem habitats due to their greater environmental heterogeneity. Residual variance, however, was substantially higher than both measured variables, explaining 70.1% of metacommunity structure.

Discussion

We tested two competing hypotheses about whether tributary junctions facilitate community connectivity or introduce ecological heterogeneities that isolate communities. Using the high throughput capabilities of eDNA metabarcoding, our research revealed diverse yet isolated invertebrate communities throughout Grand Canyon, particularly in tributaries. Invertebrate communities exhibit high beta diversity and turnover across tributary confluences and between tributary drainages, providing evidence for the influence of local habitat dynamics, rather than hydrological connectivity, on community structure. Together, these results suggest that landscape heterogeneities and flow regulation of the Colorado River restrict the distribution of biological diversity between tributary and mainstem communities in Grand Canyon. These findings support our second hypothesis that tributary communities are ecologically isolated and do not compensate for ecological degradation in the mainstem Colorado River.

eDNA detects high aquatic invertebrate biodiversity in Grand Canyon

Aquatic invertebrate communities in Grand Canyon tributaries possessed significantly higher richness compared to mainstem communities. Specifically, taxonomic richness, insect richness, and EPT richness were all significantly higher in tributaries than in the mainstem (Figure 2, Appendix S1: Figure S2, Figure S3). Tributary communities also exhibited greater variance in taxonomic richness compared to the mainstem (Figure 2), highlighting the potential influence of environmental heterogeneity in Grand Canyon tributaries. 169 taxa (25.3%) were found in both tributary and mainstem habitats, a widespread ecological discontinuity at tributary confluences where relatively few taxa are shared between mainstem and tributary habitat types.

Our eDNA sampling approach identified higher taxonomic richness in both tributary and mainstem habitats than previous studies. For example, Stevens et al. (2026) identified 72 morphospecies of aquatic invertebrates at Grand Canyon tributary confluences, compared to our identification of 667 OTUs including 161 species-level identifications. Similarly, Oberlin et al. (1999) reported 42 genera of aquatic insects in ten Grand Canyon tributaries, while we identified 604 OTUs from 171 genera across 22 tributaries, including 108 genera from the ten tributaries sampled by Oberlin et al. (1999). Cross et al. (2013) reported ~27 genera of aquatic invertebrates in the Grand Canyon segment of the Colorado River, while we identified 232 OTUs from 76 genera, including 71 species-level identifications. Notably, 74.5% of recovered OTUs in Grand Canyon communities were assigned to the order Diptera, revealing a level of taxonomic diversity previously underrepresented by studies that only report Diptera to the level of family (e.g. Cross et al., 2013; Behn and Baxter, 2019).

eDNA metabarcoding methods may be partially responsible for the findings of high taxonomic richness in Grand Canyon. Despite methodological limitations that conventionally

restrict eDNA-based community analyses to incidence-based data, eDNA analyses are effective at estimating species distributions and community structure (Goldberg et al., 2016). Paired with the increased spatial and temporal capacity of eDNA sampling (Carraro et al., 2020), our detection of taxonomically rich and diverse communities highlights eDNA metabarcoding as a tool suitable for landscape-scale community ecology in large river ecosystems.

Tributary confluences act as ecological barriers in Grand Canyon

If tributaries in the Grand Canyon river network facilitate community connectivity by linking habitats and inducing gradual shifts in habitat conditions, as predicted in our first hypothesis, we would expect high nestedness due to direct dispersal between adjacent habitats (Jones and Schmidt, 2018; Uno and Power, 2015; Wilson and McTammany, 2014). Instead, we observed few shared taxa between paired tributary and mainstem communities and high beta diversity across paired tributary and mainstem communities that was driven by turnover rather than nestedness-resultant dissimilarity (Figure 3). Low measures of nestedness-resultant dissimilarity at tributary confluences indicate that there is minimal taxonomic interchange between adjacent mainstem and tributary communities that could produce nested gradients in community composition (Miyazono and Taylor, 2013; Sasaki et al., 2012). Instead, high turnover indicates low ecological connectivity and abrupt community discontinuities between tributaries and adjacent mainstem habitats (Baselga, 2010). This pattern of high turnover may be indicative of local metacommunity dynamics (i.e., species sorting), where differences in environmental conditions promote distance and non-overlapping community structures even at short spatial scales (Baselga, 2010; Viana et al., 2016). Thus, our data does not support the hypothesis that Grand Canyon tributaries could act as a source of colonists for the river network as a whole.

The high turnover observed across tributary confluences is noteworthy given the short distances between each pair of sampling locations. At each confluence, dispersal limitations are unlikely to be a barrier to colonization, indicating instead that local environmental factors (i.e., species sorting) may be responsible for the disparity between tributary and mainstem communities (Caplat et al., 2016; Czeplédi et al., 2016; Finn and Poff, 2011). Tributary habitats in Grand Canyon have natural flow regimes with substantially lower average discharge than the Colorado River (Oberlin et al., 1999). Conditions in the mainstem Colorado River, on the other hand, are constrained by hydroelectric dam operations, which create artificial intertidal zones that reduce invertebrate diversity and selects for non-insect and generalist taxa (Abernethy et al., 2021; Blinn et al., 1995; Kennedy et al., 2016). In some ecosystems, tributaries can act as source communities or as refugia for taxa from the mainstem river (Koizumi et al., 2013; Milner et al., 2019; Pracheil et al., 2013; Troia et al., 2019; Wilson and McTammany, 2014), although few studies have explicitly studied these dynamics in unregulated tributaries of regulated mainstem river (but see Milner et al., 2019; Stevens et al., 2026). Our observations of high community dissimilarity driven by high turnover and low nestedness indicates that tributaries in Grand Canyon do not function as refugia for mainstem taxa, perhaps because generalist taxa that dominate the mainstem and are tolerant of the daily tides arising from hydropower dam operations are not adapted to the dynamic flow regimes inherent to tributaries (Mykrä and Heino, 2017). Flow and temperature regimes of the Colorado River are presently (2026) far more natural than during the time of our data collections (2022). Historically low elevations in Lake Powell Reservoir associated with a 26-year mega-drought are leading to relatively natural temperature regimes and low and relatively natural flow regimes with minimal daily fluctuations for hydropower (Palmquist et al., 2025). New research into invertebrate populations in the

Colorado River using matrix population models demonstrates that the combination of unnatural temperature and flow regimes in Grand Canyon can exclude specialist taxa such as Baetis mayflies (Kurthen et al., 2026). Thus, it seems possible that drought induced easing of these strong temperature and flow filters could eventually increase connectivity and decrease isolation among communities by allowing specialist aquatic insect taxa currently restricted to tributaries to become established in the mainstem.

Tributary communities are structured as an isolated island archipelago

Aquatic habitats throughout Grand Canyon support diverse aquatic invertebrate communities with distinct compositional differences between tributary and mainstem communities (Figure 4). These compositional differences are also revealed by the high degree of ecological network modularity and the high ratio of beta diversity attributable to species turnover. Together, these results suggest that communities in Grand Canyon are structured by local metacommunity dynamics that lead to distinct assemblages according to environmental conditions and stochastic effects (Baselga, 2010; Viana et al., 2016).

When treating tributary and mainstem communities as separate groups, multiple-site Sorensen dissimilarity was high and primarily driven by species turnover among both groups. The influence of turnover among both habitat types suggests that widespread isolation is a key influence on community structure in Grand Canyon, which is fundamentally different than most conventional river networks. In typical dendritic networks, headwater tributaries are expected to display high beta diversity due to their isolation at the extremities of the network (Campbell Grant et al., 2007; Meyer et al., 2007), but dispersal into and out of tributaries would be expected to promote patterns of nestedness in mainstem habitats through mass effects, refugia dynamics,

and the accumulation of biodiversity along a downstream gradient (Heino et al., 2015; Uno and Power, 2015). In Grand Canyon, however, both tributary and mainstem communities exhibit high turnover and low nestedness-resultant dissimilarity. This pattern is indicative of strong environmental or biotic filters operating across short dispersal distances, creating a discontinuous community structure that resembles an island archipelago with unique environmental characteristics and limited ecological connectivity (Aguilée et al., 2021; MacArthur and Wilson, 1963; Whittaker et al., 2017).

Community diversity among habitats in the mainstem Colorado River was driven by high turnover and low nestedness. In mainstem rivers, aquatic communities could be expected to display patterns of nestedness when gradual downstream changes in environmental characteristics are overlain with successive tributaries that contribute biodiversity and potential new colonists downstream (Bruns et al., 1984; Jones and Schmidt, 2018, 2017). This pattern should be particularly strong in rivers affected by impoundment and other anthropogenic alteration, where ecosystem recovery occurs gradually in the downstream direction and is punctuated by occasional tributary inputs as well (Abernethy et al., 2021; Ward and Stanford, 1983). The canyon-bound geography of Grand Canyon and the small relative size of tributaries compared to the mainstem Colorado River, however, reduce the importance of successive tributary inputs on invertebrate communities (Jones and Schmidt, 2017), thereby contributing to the pattern of low nestedness observed in Grand Canyon aquatic ecosystem. In the absence of nestedness playing a strong role, species turnover arising from environmental filters and stochastic processes appears to be the primary influence on community diversity in the mainstem Colorado River.

Fragmented communities with decreased connectivity among habitats are less resilient to and slower to recover from disturbances such as floods and droughts (Merriam and Petty, 2016). dbRDA-based variance partitioning found that environmental variability among tributary habitats explained a significantly greater proportion of community structure than spatial variation along the river network (Table 1). Several of the significant environmental variables are potentially related to ecosystem disturbances like flooding, including substrate particle size and percent barren terrain. Furthermore, the large proportion of unexplained residual variance, a common feature of aquatic invertebrate community analyses (Ge et al., 2021; Schmera et al., 2018), may reflect unmeasured cycles of disturbance and colonization that often shape metacommunity organization in arid river ecosystems (Lytle, 2001; Lytle and Poff, 2004; Townsend, 1989). The isolated nature of Grand Canyon tributary communities contributes to the high invertebrate biodiversity observed in this system (Schmera et al., 2018), yet this isolation also increases the likelihood that local stochastic disturbances might reduce local diversity and limit the resilience and speed at which communities recover (Merriam and Petty, 2016).

Conclusion

In summary, we found that aquatic invertebrate communities in Grand Canyon are structured like island archipelagos, with tributary communities isolated from each other and from proximate habitats in the mainstem Colorado River. Environmental heterogeneity and stochastic processes in general, and large hydrological discontinuities at tributary confluences in particular, appear to promote distinct community structure across adjacent habitats in Grand Canyon. Community isolation therefore drives high overall biodiversity in Grand Canyon, yet this isolation leaves these communities vulnerable to stochastic disturbances and anthropogenic

effects. Metacommunity dynamics of species sorting and dispersal limitations reflect strong landscape structure that influences the distribution of species between habitats (Suzuki and Economo, 2021; Vellend, 2010). Management of communities in heterogeneous landscapes, such as aquatic invertebrates in Grand Canyon and other desert river networks, may benefit from a holistic view of the landscape heterogeneities that promote unique and biodiverse communities.

Acknowledgements

We thank Kate Behn, Ann-Marie Bringham, Anya Metcalfe, Nick Paretti, and Jessica Anderson for their assistance with fieldwork logistics and sampling, as well as Chris Hunsaker and other river guides from Ceiba for safely navigating us through Grand Canyon. Thank you to Angelika Kurthen, Isaac Shuman, Aaron Klarenbach, and other members of the Lytle Lab at Oregon State University for guidance with sample collection, data interpretation and reviewing drafts. Funding for this study came from the Bureau of Reclamation's Glen Canyon Dam Adaptive Management Program and a grant from the US Geological Survey/National Park Service – Water Quality Partnership program. Any use of trade, product, or firm names is for descriptive purposes only and does not imply endorsement by the US government.

Conflict of Interest Statement

The authors declare no conflicts of interest.

Author Contributions

Jared W. Freedman: Conceptualization, field sample collection, laboratory sample processing, analysis, writing – original draft preparation and editing. **Theodore A. Kennedy:**

553 Conceptualization, field sample collection, writing – review and editing, funding acquisition.
554 **Molly K. Burke:** Molecular lab methodology, writing – review and editing. **David A. Lytle:**
555 Conceptualization, field sampling, analysis, writing – review and editing.

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837

838 **Tables**

839 *Table 1.* dbRDA-based variance partitioning of environmental, spatial, shared, and residual
 840 variance of aquatic invertebrate community structure in Grand Canyon tributaries.

<i>Component of Variance</i>	<i>Adj. R²</i>	<i>Pseudo-F</i>	<i>p</i>	<i>Variables Included</i>
				Temperature, drainage relief, D15
<i>Environmental</i>	0.196	2.771	<0.0001	substrate, and drainage percent barren terrain
<i>Spatial</i>	0.047	3.405	<0.0001	River network distance
<i>Shared</i>	0.056			
<i>Residual</i>	0.701			

841

Figure captions

Figure 1. Map of eDNA sampling location in Grand Canyon. eDNA samples were collected in spring 2022 from perennial Grand Canyon tributaries (blue circles) and from the mainstem Colorado River directly upstream from each tributary confluence (orange circles). The mainstem Colorado River is highlighted with a thicker line than perennial tributaries. Inset shows location of study area relative to the contiguous United States. Photograph panels on the right display the confluence of Tapeats Creek (river km 216.3; top) and Clear Creek (river km 136.3; bottom). Photographs by Jared Freedman.

Figure 2. A. Observed OTU richness of Grand Canyon tributary and mainstem communities. Horizontal bars represent sites, and stacked colors represent taxonomic order assigned to OTUs. See Appendix S1 (Table S1) for explanation of site abbreviations. The majority of taxa identified with eDNA were Diptera, highlighting an underappreciated aspect of aquatic invertebrate biodiversity in Grand Canyon. **B.** Box plot of OTU richness of aquatic invertebrate communities in mainstem and tributary sites. Tributaries had significantly higher OTU richness than the mainstem (Wilcoxon $W = 471$, $p < 0.0001$, $r = 0.71$).

Figure 3. Mean partitioned beta diversity for paired tributary and mainstem sites in Grand Canyon. Sorensen dissimilarity was high between paired sites (mean $\beta_{\text{sor}} = 0.695$), and spatial turnover (mean $\beta_{\text{sim}} = 0.516$) was a significantly greater component of community diversity than nestedness (mean $\beta_{\text{sne}} = 0.178$; Wilcoxon signed-rank $V = 249$, $p < 0.001$). High community dissimilarity and turnover between adjacent tributary and mainstem communities is suggestive of an ecological discontinuity at tributary confluences in Grand Canyon.

Figure 4. Nonmetric multidimensional scaling ordinations (metric $R^2 = 0.881$, stress = 0.1239) show strong dissimilarity between tributary and mainstem aquatic insect communities. Points

865 represent Grand Canyon aquatic invertebrate communities in species space, with distance
866 between points indicating community dissimilarity. Square points represent mainstem
867 communities and round points represent tributary communities. Tributary communities were
868 distinct from one another and from the mainstem, while the mainstem communities were more
869 similar to one another (PERMANOVA $R^2 = 0.250$, $p < 0.0001$; PERMDISP $p < 0.0001$). Both
870 tributary and mainstem communities exhibit significant shifts in structure as a function of
871 distance from Glen Canyon Dam (PERMANOVA $R^2 = 0.066$, $p = 0.0051$).